THE PATENTS ACT, 1970

(39 OF 1970)

AND THE PATENTS RULES, 2003

FORM 1

(See Section 7, 54 and 135 and Rule 20(1))

APPLICATION FOR GRANT OF PATENT

No.	Field	Details
1.	Applicant's Name	[FULL LEGAL NAME OF APPLICANT / INVENTOR]
2.	Nationality	Indian National
3.	Address	[Complete Postal Address], [City], [State], PIN: [XXXXXX], India
4.	Name of Inventor	[FULL LEGAL NAME] — Same as Applicant
5.	Address of Inventor	Same as above
6.	Title of Invention	A MULTI-DOMAIN CYBER-PHYSICAL INTRUSION DETECTION AND RESPONSE SYSTEM FOR ELECTRIC POWER DISTRIBUTION NETWORKS USING PHYSICS-INFORMED NEURAL NETWORKS AND MULTI-LAYER ANOMALY FUSION
7.	Priority Document	Provisional Application No. ___/___/2026 dated ___/___/2026
8.	Divisional of	Not Applicable
9.	Convention Application	Not Applicable
10.	Patent of Addition to	Not Applicable

I/We request that a patent may be granted to me/us for the invention described in the Complete Specification accompanying this application.

Signature: _____ Date: _____

Name: [FULL LEGAL NAME]

Address: [Complete Address], India

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Application No.: _____ Date of Filing: _____ Filing Fee Receipt No.: _____

THE PATENTS ACT, 1970

FORM 2

(See Section 10 and Rule 13)

COMPLETE SPECIFICATION

TITLE OF THE INVENTION

A MULTI-DOMAIN CYBER-PHYSICAL INTRUSION DETECTION AND RESPONSE SYSTEM FOR ELECTRIC POWER DISTRIBUTION NETWORKS USING PHYSICS-INFORMED NEURAL NETWORKS AND MULTI-LAYER ANOMALY FUSION

1. FIELD OF THE INVENTION

The present invention relates to apparatus and methods for securing electric power distribution networks against cyber-physical attacks. More specifically, the invention provides a system comprising four cooperating hardware and software engines that together monitor the cyber, electrical-physics, and thermodynamic domains of an operational power distribution network simultaneously, producing the technical effect of detecting, classifying, and automatically responding to intrusion attempts — including False Data Injection attacks, electricity theft via smart meter tampering, unauthorised SCADA command injection, industrial device impersonation, coordinated load manipulation, and replay attacks — that are undetectable by any system confined to network-layer monitoring alone. The invention further provides real-time voltage stability monitoring and automated incident reporting conforming to national cybersecurity reporting obligations.

2. BACKGROUND OF THE INVENTION

2.1 The Cyber-Physical Nature of Modern Power Distribution Networks

Electric power distribution networks have evolved from electrically isolated analogue systems into fully integrated cyber-physical systems in which digital commands and sensor data streams produce direct, immediate physical consequences. The widespread deployment of SCADA (Supervisory Control and Data Acquisition) systems, Programmable Logic Controllers (PLCs), Remote Terminal Units (RTUs), Intelligent Electronic Devices (IEDs), and Advanced Metering Infrastructure (AMI) has introduced

pervasive digital attack surfaces into critical infrastructure that, if compromised, results in physical harms including grid failures, equipment damage, protection relay miscoordination, and cascading blackouts affecting millions of consumers.

India's electric power sector presents a uniquely severe instance of this challenge. The Revamped Distribution Sector Scheme (RDSS), mandated by the Government of India, requires deployment of two hundred and fifty million smart meters by 2026, creating the world's largest single expansion of cyber-physical attack surface in a power distribution context. Data published by the Indian Computer Emergency Response Team (CERT-In) documents a two hundred and seventy-eight percent increase in cyber incidents targeting Indian power infrastructure between 2018 and 2022. Threat intelligence published by Recorded Future's Insikt Group confirmed sustained intrusions by a nation-state Advanced Persistent Threat group into seven Indian State Load Despatch Centres (SLDCs) during 2021. During the India-Pakistan conflict of 2025, the Union Minister of Power confirmed more than two lakh cyberattacks directed at Indian power sector infrastructure within a single engagement period. Non-Technical Losses attributable to electricity theft cost Indian distribution utilities an estimated twenty thousand crore rupees annually, the substantial majority of which involves smart meter hardware and firmware tampering that is invisible to all currently deployed monitoring systems.

2.2 The Fundamental Inadequacy of Network-Layer Intrusion Detection Systems

The prior art in industrial control system security comprises intrusion detection systems that monitor industrial network protocol traffic — primarily Modbus TCP, DNP3, IEC-61850, and OPC-UA — for protocol anomalies, unauthorised commands, and device behavioural deviations. These systems, including the commercial platforms disclosed in US Patent Nos. 10,609,068; 11,223,637; and 10,764,316, and the academic systems described in prior literature, operate exclusively in the cyber domain. They have no technical capability to validate whether reported sensor measurements are physically consistent with the laws governing the electrical operation of the monitored network.

This limitation has a formal mathematical basis established by Liu, Ning, and Reiter (ACM Conference on Computer and Communications Security, 2011). Their analysis proves that an attacker constructing a False Data Injection attack vector c satisfying the linear relationship:

$$c = H * a$$

where H is the linearised measurement Jacobian of the power network state estimation model and a is any arbitrary non-zero vector, produces sensor manipulations that are completely undetectable by any chi-squared residual test applied to network-layer data. This is not a practical limitation of existing systems — it is a mathematical proof of impossibility. Network-layer IDS cannot, even in principle, detect FDI attacks crafted to satisfy this condition. The attacker who knows the grid topology can manipulate voltage, power, and current sensor readings to cause a state estimator to compute a fundamentally incorrect grid operating point — triggering incorrect protection relay operations, incorrect load dispatch, and potentially cascading failures — while every network monitoring tool reports entirely normal operation.

The same fundamental limitation applies to electricity theft detection. A consumer who bypasses a smart meter or shunts a current transformer draws real power from the distribution transformer but reports zero or reduced consumption to the metering system. The discrepancy between actual feeder power injection and the sum of metered consumption is a physical energy balance violation — detectable only by comparing measurements at two physically separate points in the distribution network. This

cross-domain comparison is unavailable to any system confined to monitoring the communication network.

2.3 Specific Limitations of the Prior Art

Existing intrusion detection systems for power sector industrial control systems suffer from one or more of the following technical limitations that the present invention is designed to overcome:

(a) Confinement to the cyber domain: all commercially deployed OT/ICS security platforms, including market-leading products, perform no validation of sensor measurements against the physical laws of the power network. They are therefore incapable of detecting FDI attacks, electricity theft, or any attack that manipulates sensor readings while maintaining network-layer protocol compliance.

(b) Dependence on linearised grid models: prior academic approaches that do incorporate physics-based detection (Pasqualetti et al., 2013; Bobba et al., 2010) rely on linearised DC power flow approximations or linearised state estimation Jacobians, which are provably bypassable by the attack vector $c = Ha$. The present invention enforces the full nonlinear AC power flow equations, which cannot be satisfied by a linearised bypass vector.

(c) Absence of uncertainty quantification: prior art systems provide deterministic anomaly scores without confidence intervals, producing elevated false positive rates under legitimate load transients — diurnal demand cycles, sudden load switching, and reactive power compensation events — that cause voltage and power deviations superficially resembling attack signatures.

(d) Single-domain detection: no prior art system simultaneously monitors the cyber domain, the electrical physics domain, and the thermodynamic side-channel domain, depriving the detection system of the corroborating evidence that enables confident threat classification and automated response with reduced false positive rate.

(e) Absence of automated, physics-informed response: prior art generates alerts but does not implement automated, physically-grounded mitigation actions informed by the physics of the specific attack. A response to an FDI attack requires different physical actions than a response to a SCADA command injection or an electricity theft event.

(f) Inability to distinguish cyber attack from physical fault: prior art does not implement causal analysis to distinguish a cyber attack originating at a specific attacker entry point from a hardware sensor failure or a physical distribution fault — each requiring fundamentally different response actions.

3. OBJECTS OF THE INVENTION

The primary objects of the present invention are:

(i) To provide a system that produces the technical effect of protecting a physical electric power distribution network from False Data Injection attacks by detecting physically inconsistent sensor readings before they cause incorrect protective relay operation, incorrect load dispatch, or cascading grid failures.

(ii) To provide a system that produces the technical effect of detecting electricity theft in a smart-metered distribution network through feeder-level energy balance computation that is immune to individual smart meter tampering, thereby reducing Non-Technical Losses.

(iii) To provide a system that detects malicious commands injected into SCADA communication networks — including unauthorised circuit breaker operations, protection relay setting changes, and load setpoint manipulations — before they produce harmful physical effects on the distribution network.

(iv) To provide a Physics-Informed Neural Network trained with a composite loss function that simultaneously enforces data fidelity and full nonlinear AC power flow governing equations, producing physics residuals that detect attacks bypassing linearised chi-squared detection methods.

(v) To provide a multi-domain anomaly fusion engine that combines physics-domain and network-domain detection signals with confidence-weighted adaptive scoring, producing a unified threat level with corroboration-amplified confidence when multiple independent physical domains simultaneously detect anomalous conditions.

(vi) To provide explainable alert generation that identifies the specific bus, measurement variable, physical law, and attack type responsible for each detected event, enabling grid operators to take targeted physical protective actions.

(vii) To provide automated generation of incident reports conforming to CERT-In mandatory notification requirements, reducing operator workload during active attack scenarios.

(viii) To provide a system deployable in both cloud-connected and fully offline edge-computing configurations, enabling operation at remote rural substations lacking reliable network connectivity.

(ix) To provide a device identity authentication mechanism based on hardware-level response timing fingerprinting that detects device impersonation attacks invisible to protocol-layer analysis.

(x) To provide a causal root-cause analysis mechanism that distinguishes cyber attacks from physical equipment faults and sensor hardware failures, preventing costly unnecessary security response to non-attack anomalies.

4. SUMMARY OF THE INVENTION

According to a first aspect of the present invention, there is provided a cyber-physical intrusion detection and response system for securing an electric power distribution network, comprising: a first detection engine configured to monitor industrial network protocol traffic and detect anomalies in the cyber domain; a second detection engine configured to validate observed sensor measurements against expected values derived from a Physics-Informed Neural Network and an AC power flow physics oracle, detecting anomalies in the electrical physics domain; a third detection engine configured to monitor hardware thermal and electromagnetic emission signatures, detecting anomalies in the thermodynamic domain; and a fusion and response engine configured to combine signals from said detection engines into a unified threat score and execute automated protective response actions on the power distribution network.

According to a second aspect of the present invention, there is provided a computer-implemented method for protecting a physical electric power distribution network from False Data Injection attacks, said method producing the technical effect of detecting physically inconsistent sensor measurements before they cause erroneous state estimation and incorrect protective relay operation, the method comprising: receiving observed sensor readings; computing a physics-consistent expected state using a Physics-Informed Neural Network trained with a composite loss enforcing AC power flow equations; computing a FDI residual; applying a statistical threshold; and generating an actionable alert identifying the affected network bus and physical law violated.

According to a third aspect of the present invention, there is provided a computer-implemented method for detecting electricity theft in a physical smart-metered distribution network, said method producing the technical effect of recovering non-technical energy losses through detection of energy balance violations at the distribution feeder level, the method comprising: receiving feeder-level power injection and individual meter readings; computing a Non-Technical Loss ratio independent of individual meter integrity; and generating an alert identifying the feeder segment and estimated magnitude of theft.

According to a fourth aspect of the present invention, there is provided a method for fusing anomaly signals from multiple independent physical detection domains into a unified threat level for a cyber-physical power distribution network, said method producing the technical effect of reducing false positive rates through independent cross-domain corroboration while maintaining high detection sensitivity for sophisticated coordinated attacks.

5. BRIEF DESCRIPTION OF THE DRAWINGS

The invention will be better understood from the following description taken in conjunction with the accompanying drawings, wherein:

FIG. 1 is a high-level system architecture diagram illustrating the four principal detection engines of the present invention — Dvvalin (10), AEthelred (20), Brynhildr (30), and Valkyrie (40) — together with their respective monitored physical domains (cyber, electrical physics, and thermodynamic) and the data flows between them.

FIG. 2 is a detailed block diagram illustrating the six-module detection pipeline of the AEthelred engine (20), showing: Module 1 Data Ingestion (21), Module 2

Physics Constraints Engine (22), Module 3 Physics-Informed Neural Network (23), Module 4 Anomaly Detection (24), Module 5 Threat Identification (25), and Module 6 Response and Visualisation (26), with data flow arrows connecting each module.

FIG. 3 is a schematic diagram illustrating the Physics-Informed Neural Network (PINN) architecture of Module 3 (23), showing the input layer receiving normalised bus identifier and timestamp, five hidden layers with hyperbolic tangent activation, the output layer producing predicted voltage magnitude, voltage angle, active power and reactive power, and the composite training loss function L_total comprising three additive loss terms.

FIG. 4 is a block diagram illustrating the Valkyrie multi-domain fusion engine (40) showing confidence-weighted score computation for physics-domain score S_A and network-domain score S_D, normalisation, weighted combination, corroboration bonus application, and threshold-based threat level classification.

FIG. 5 is a graph illustrating physics residual magnitude as a function of time, showing the baseline residual distribution during normal grid operation, the residual spike onset upon injection of a False Data Injection attack at t=0, and the anomaly detection threshold tau, demonstrating a residual amplification factor of approximately 157-fold relative to the normal baseline.

FIG. 6 is a graph illustrating Non-Technical Loss ratio over time showing the baseline NTL ratio during normal metered operation, the step increase upon initiation of electricity theft simulation at bus 7, and the NTL detection threshold of 0.05 (5%), demonstrating detection within the slow-window detection period.

FIG. 7 is a block diagram of the three deployment configurations supported by the present invention: cloud-connected configuration, edge-deployed offline configuration, and federated multi-substation configuration.

FIG. 8 is a schematic diagram illustrating the Graph Attention Network variant of Module 3 (23), showing the power network represented as a graph with buses as nodes and transmission lines as edges, with attention weight computation between neighbouring nodes and physics residual loss aggregated across the graph.

6. DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The following description sets forth specific details of preferred and alternative embodiments of the invention. It will be apparent to those skilled in the relevant arts that modifications, variations, and equivalents are possible without departing from the spirit and scope of the invention as defined by the appended claims. The description and accompanying examples are not to be construed as limiting. All such modifications and equivalents falling within the scope of the appended claims are intended to be covered.

6.1 System Overview and Core Inventive Principle

Referring to FIG. 1, the invention in its preferred embodiment comprises four cooperating detection engines, each monitoring a distinct physical domain of the target power distribution network. The four engines are designated as follows in this specification:

(a) The SCADA Network Intrusion Detection Engine, designated Dvvalin (10), monitors the cyber domain by analysing industrial network protocol traffic.

(b) The Physics-Informed Anomaly Detection Engine, designated AEthelred (20), monitors the electrical physics domain by validating sensor measurements against physical laws governing power flow in the network.

(c) The Thermodynamic Side-Channel Security Engine, designated Brynhildr (30), monitors the thermodynamic domain by analysing hardware thermal and electromagnetic emission signatures of field devices.

(d) The Multi-Domain Fusion and Response Engine, designated Valkyrie (40), combines anomaly signals from Dvvalin (10), AEthelred (20), and Brynhildr (30) into a unified threat score and drives automated protective response actions on the physical distribution network.

Core Inventive Principle: *An attacker cannot simultaneously make network traffic appear legitimate to Dvvalin (10), inject sensor values satisfying the full nonlinear AC power flow equations monitored by AEthelred (20), AND maintain thermodynamically consistent hardware signatures monitored by Brynhildr (30) without leaving detectable inconsistencies across at least one domain. This multi-domain approach provides a security guarantee that is qualitatively stronger than any single-domain intrusion detection system.*

6.2 AEthelred — Physics-Informed Anomaly Detection Engine (20)

Referring to FIG. 2, the Physics-Informed Anomaly Detection Engine (20) implements a six-module detection pipeline. Each module has a defined input, processing function, and output contract, enabling modular replacement and extension.

6.2.1 Module 1 — Data Ingestion and Protocol Semantic Mapping (21)

Module 1 (21) receives raw industrial network packets captured at a network monitoring point — in the preferred embodiment, a SPAN port or network tap on the substation LAN — and extracts the following physical state variables from protocol payloads:

(i) Voltage magnitude V_i at bus i, expressed in per-unit (pu) normalised to the nominal system voltage.

(ii) Voltage phase angle θ_i at bus i, expressed in degrees.

(iii) Active power injection P_i at bus i, expressed in megawatts (MW).

(iv) Reactive power injection Q_i at bus i, expressed in megavolt-amperes reactive (MVAr).

(v) System frequency f, expressed in hertz (Hz).

Module 1 supports the following industrial protocols in the preferred embodiment: Modbus TCP (TCP port 502); DNP3 (TCP/UDP port 20000); IEC-61850 GOOSE (Ethernet EtherType 0x88B8); IEC-61850 MMS (TCP port 102); and OPC-UA (TCP port 4840). Protocol parsing is implemented using the Zeek network analysis framework with custom industrial protocol scripts that extract register values and map them to the physical state variables listed above according to a configurable register-to-variable mapping table.

The output of Module 1 is a time-indexed stream of Physical State Reading objects, each comprising: a timestamp in Unix epoch format; a bus identifier; and the extracted values for V_i, θ_i, P_i, Q_i, and f. This stream is supplied concurrently to Module 3 and Module 4 of the AEthelred pipeline.

6.2.2 Module 2 — Physics Constraints Ingestion and Anchoring (22)

Module 2 (22) computes and maintains the physics-consistent expected state of the network by solving the full, nonlinear AC power flow equations for the network topology. In the preferred embodiment, Module 2 employs the PyPSA (Python for Power System Analysis) framework, which solves the Newton-Raphson power flow algorithm for the full complex admittance model of the network.

For a network of N buses, Module 2 computes the following for each bus i:

$$P_i = \sum_{j=1}^{N} V_i * V_j * (G_{ij} * \cos(\theta_i - \theta_j) + B_{ij} * \sin(\theta_i - \theta_j)) \quad ...(1)$$

$$Q_i = \sum_{j=1}^{N} V_i * V_j * (G_{ij} * \sin(\theta_i - \theta_j) - B_{ij} * \cos(\theta_i - \theta_j)) \quad ...(2)$$

where V_i and V_j are the voltage magnitudes at buses i and j respectively; θ_i and θ_j are the voltage phase angles at buses i and j; G_{ij} is the real component (conductance) of the complex admittance matrix element Y_{ij}; and B_{ij} is the imaginary component (susceptance) of Y_{ij}. These computations constitute the physics oracle function $h(x_{expected})$ which represents the physically correct expected sensor readings for the network under the current estimated load conditions. These values are stored as the Module 2 output and are periodically updated as load conditions change.

Module 2 additionally computes the following quantities supplied to Module 4:

(a) The Kirchhoff Current Law (KCL) residual at each bus i, computed as the magnitude of the net complex current injection, which should be zero under balanced power flow conditions.

(b) The Non-Technical Loss (NTL) ratio for electricity theft detection:

$$NTL_ratio = (P_feeder_in - SUM_{k=1}^{M} P_meter_k - P_estimated_loss) / P_feeder_in \quad ...(3)$$

where P_feeder_in is the power entering the feeder at the substation metering point; P_meter_k is the consumption reading from the k-th smart meter among M meters on the feeder; and P_estimated_loss is the technically estimated resistive line loss computed from the feeder impedance model and measured current. An NTL_ratio exceeding the configured threshold (default 0.05, representing 5 percent) indicates electricity theft or widespread meter tampering. This computation is mathematically independent of the integrity of any individual meter and cannot be defeated by manipulating individual meter firmware or hardware.

(c) The voltage stability L-index at each load bus j:

$$L_j = | 1 + SUM_{i \text{ in } generator_buses} (F_ji * V_i / V_j) | \quad ...(4)$$

$$\text{where } F = - inv(Y_LL) * Y_LG \quad \text{(Thevenin equivalent matrix)} \quad ...(5)$$

where Y_LL is the submatrix of the admittance matrix corresponding to load bus interconnections and Y_LG is the submatrix corresponding to load-to-generator bus connections. The L-index value of unity indicates incipient voltage collapse. Values exceeding 0.5 indicate a high-risk voltage stability condition warranting operational intervention.

6.2.3 Module 3 — Physics-Informed Neural Network Prediction Model (23)

Module 3 (23) implements a Physics-Informed Neural Network (PINN) that predicts the physics-consistent expected physical state of the network as a function of bus identifier and time. The PINN is trained exclusively on data collected during confirmed normal operation of the target network, with no labelled attack data required.

The PINN is trained with a composite loss function that simultaneously enforces data fidelity with observed normal-operation measurements and adherence to the governing PDEs of the power flow system. The general form of the composite loss is:

$$L_total = lambda_d * L_data + lambda_p * L_physics + lambda_b * L_boundary \quad ...(6)$$

$$L_data = (1 / N_d) * SUM_{i=1}^{N_d} || u_hat(x_i, t_i) - u_i ||^2 \quad ...(7)$$

$$L_physics = (1 / N_r) * SUM_{j=1}^{N_r} || N[u_hat](x_j, t_j) - f(x_j, t_j) ||^2 \quad ...(8)$$

*L_boundary = (1 / N_b) * SUM_{k=1}^{N_b} || B[u_hat](x_k, t_k) - g(x_k, t_k) ||^2 ...(9)*

where u_hat(x,t) is the PINN network output representing the predicted physical state; u_i are the observed sensor measurements from Module 1 during normal operation training; N[u_hat] is the physics differential operator applied to the PINN output, specifically encoding the AC power flow residual (the difference between the network-injected power and the power computed from Kirchhoff's laws); f(x,t) is the corresponding source term; B[u_hat] encodes boundary conditions (slack bus reference voltage and angle, feeder endpoint conditions); g(x,t) are the boundary values; and lambda_d, lambda_p, lambda_b are weighting coefficients (preferred default values: lambda_d = 1.0, lambda_p = 0.5, lambda_b = 1.0).

The physics operator N embeds Kirchhoff's Current Law and the AC power flow equations (1) and (2) as differentiable constraints. During training on normal grid data, the physics residual term L_physics converges to a small value (in the preferred embodiment, below 0.001 in normalised units) because normal grid operation obeys these laws. At inference time, when an attacker injects false sensor values that do not satisfy the AC power flow equations, L_physics rises sharply, providing the primary anomaly detection signal.

In the preferred embodiment, the PINN architecture comprises:

(i) An input layer receiving a two-dimensional input vector comprising the normalised bus identifier (bus index divided by total bus count) and the normalised timestamp.

(ii) Five hidden layers each comprising sixty-four neurons with hyperbolic tangent activation functions.

(iii) An output layer of four neurons producing the predicted normalised voltage magnitude, normalised voltage angle, normalised active power, and normalised reactive power.

The PINN is implemented using the DeepXDE library with a PyTorch computational backend and trained using the Adam optimiser with learning rate decay.

In a first alternative embodiment, Module 3 implements a Physics-Informed Graph Attention Network (PI-GAT) wherein the power network is represented as a directed graph G = (V, E) with buses as nodes V and transmission lines as edges E. The Graph Attention Network learns to aggregate neighbourhood information using attention-weighted message passing, enabling the physics residual computation to exploit topological relationships between electrically adjacent buses. An FDI attack affecting bus i propagates its inconsistency to the attention representations of neighbouring buses i-1, i+1, and i+2, producing a characteristic graph-level residual pattern that improves detection sensitivity by approximately ten to fifteen percent compared to the feedforward PINN embodiment.

In a second alternative embodiment, Module 3 implements a Bayesian Physics-Informed Neural Network using Monte Carlo Dropout. Dropout layers with probability p (preferred value: 0.10) are applied between all hidden layers and remain active during inference. N stochastic forward passes (preferred N = 100) are performed for each inference query, producing a predictive mean mu_hat(x,t) and a per-output predictive standard deviation sigma_hat(x,t). The anomaly detection criterion in Module 4 is then modified to trigger only when the observed measurement deviates from the predicted mean by more than a

configurable multiple of the predicted standard deviation (preferred threshold: 3.0 standard deviations). This uncertainty-aware detection mechanism substantially reduces false positive rates under normal load transients.

6.2.4 Module 4 — Real-Time Anomaly Detection (24)

Module 4 (24) receives the observed physical state readings from Module 1 and the expected physical state values from Module 3 (PINN prediction) and Module 2 (physics oracle), and computes the False Data Injection residual:

r(z) = | z_observed - h(x_expected) | ...(10)

*h(x_expected) = alpha * u_hat_PINN(x, t) + (1 - alpha) * h_PyPSA(x) ...(11)*

where z_observed is the observed sensor reading from Module 1; u_hat_PINN(x,t) is the PINN prediction from Module 3; h_PyPSA(x) is the AC power flow oracle value from Module 2; and alpha is a configurable weighting coefficient (preferred default: 0.5). The dual-oracle formulation of h(x_expected) requires an attacker to simultaneously deceive both the learned PINN and the analytically computed physics oracle — a substantially harder evasion requirement than deceiving either alone.

A statistical anomaly score is then computed by applying a regression model f to the residual:

X = f(r(z)) [preferred: Random Cut Forest or Isolation Forest] ...(12)

V = X^2 > tau [anomaly flag; tau calibrated by Neyman-Pearson criterion] ...(13)

Additionally, Module 4 applies a classical chi-squared residual test:

|| r(z) ||^2 > chi^2_{alpha}(m) [chi-squared with m measurements, level alpha] ...(14)

The PINN-based residual (equations 10-13) detects sophisticated FDI attacks that satisfy the linearised chi-squared test — the c = Ha bypass condition of Liu et al. (2011) — because the PINN enforces the full nonlinear AC power flow equations (1) and (2), which cannot be simultaneously satisfied by the linearised bypass vector a. This is a key technical advantage of the present invention over all prior art.

Module 4 further implements a multi-timescale detection architecture:

(a) Fast detection window: a sliding window of N_fast consecutive readings (preferred default: 3 readings at 2-second polling intervals, covering 6 seconds), computing the maximum physics residual within the window. This window is sensitive to transient FDI attacks affecting a single or small number of readings.

(b) Slow detection window: a sliding window of N_slow consecutive readings (preferred default: 150 readings at 2-second intervals, covering 5 minutes), computing the 90th percentile of the NTL ratio within the window. This window

accumulates evidence of gradual electricity theft and slow-ramp load manipulation attacks that are not detectable in individual snapshots.

The combined anomaly score from Module 4 is the maximum of scores from both detection windows.

6.2.5 Module 5 — Intrusion Identification and Classification (25)

Module 5 (25) receives confirmed anomaly events from Module 4 and performs threat classification, root-cause analysis, and SNR-based feature filtering.

Feature vectors for classification are pre-processed by SNR-based feature selection, wherein features are ranked by the ratio of between-class variance to within-class variance (equivalent to the F-statistic of a one-way ANOVA):

$$SNR_feat(f_i) = Var_between(f_i) / Var_within(f_i) \quad ...(15)$$

Features with SNR below a configured threshold are excluded from the classification input, and training instances more than 1.5 interquartile ranges from the feature quartiles are removed by IQR filtering. This pre-processing maximises the effective signal-to-noise ratio of the classification model input.

Module 5 performs causal root-cause analysis using Granger causality testing on residual time series to distinguish three anomaly categories:

(i) Cyber attack: the anomaly exhibits causal propagation from a specific source bus to neighbouring buses, consistent with an attacker's entry point.

(ii) Physical equipment fault: the anomaly appears simultaneously across electrically coupled buses without a single causal source, consistent with a physical fault event.

(iii) Sensor hardware failure: the anomaly appears in isolation at a single bus with no spatial correlation to neighbouring buses, consistent with a hardware failure at that measurement point.

This classification prevents unnecessary cybersecurity response to physical equipment faults and sensor failures, reducing operational disruption.

6.2.6 Module 6 — Response, Mitigation, and Visualisation (26)

Module 6 (26) receives classified threat events from Module 5 and executes the following response and documentation actions:

(a) MITRE ATT&CK for ICS mapping: each classified threat event is automatically mapped to the corresponding tactic and technique in the MITRE ATT&CK for Industrial Control Systems framework (e.g., FDI → T0832 Manipulation of View; SCADA command injection → T0855 Unauthorized Command Message; device spoofing → T0886 Remote Services).

(b) SHAP explainability: SHAP (SHapley Additive exPlanations) values are computed for each alert, decomposing the anomaly score into per-feature contributions and identifying the dominant physical variable and bus location responsible for the alert.

(c) Network-layer response: anomaly scores are translated into Suricata network IDS rules via Lua script integration, enabling network-layer protective actions including connection blocking and traffic rate limiting.

(d) SIEM logging: all events are streamed to Wazuh / OpenSearch with a tamper-evident hash-chain audit log, wherein each entry contains the cryptographic hash of the preceding entry, satisfying evidence preservation requirements under Section 79A of the Information Technology Act, 2000.

(e) Incident report generation: for threat events at HIGH or CRITICAL threat level, a structured incident report is automatically generated conforming to the CERT-In mandatory notification format, including attack type, timeline, affected components, MITRE mapping, and recommended actions.

(f) Operator notification: configurable alerts are dispatched to designated operators via SMS and messaging platform APIs.

6.3 Dvvalin — SCADA Network Intrusion Detection Engine (10)

The SCADA Network Intrusion Detection Engine (10) monitors industrial network traffic and produces a network-domain anomaly score S_D by implementing the following detection layers:

6.3.1 Rule-Based Detection Layer

A rule-based detection layer applies deterministic, stateless rules to each incoming industrial protocol packet, flagging known attack patterns with a rule confidence score of 0.9 or above. Rules in the preferred embodiment include:

(i) Write commands (Modbus function codes 5, 6, 15, 16; DNP3 function codes 3, 4) directed to critical device registers on PLCs or IEDs designated as write-protected in the system configuration.

(ii) Protocol function codes outside the set of codes observed during the normal operation baseline period — indicative of firmware probing or exploit delivery.

(iii) Broadcast-addressed Modbus requests (Unit ID 0xFF) — indicative of device enumeration reconnaissance.

(iv) Malformed packet headers where the declared protocol identifier or length field is inconsistent with the actual packet structure.

6.3.2 Per-Device Behavioural Profiling

Each ordered pair of communicating devices (source device, destination device) is assigned a behavioural profile during a configurable training period of normal operation. The profile records: the empirical frequency distribution of function codes observed; the range of register addresses accessed; the distribution of inter-packet arrival times; and the set of register addresses accessed. Deviations from the trained profile are scored using a One-Class Support Vector Machine with radial basis function kernel.

6.3.3 Device Timing Fingerprinting

Every physical industrial controller exhibits a characteristic distribution of response latency times determined by its firmware version, processor speed, and hardware design. Engine (10) records the response time distribution for each device identifier during the training period, fitting a normal distribution with parameters (mu_device, sigma_device). At inference time, for each batch of response times observed for a claimed device identifier, a Kolmogorov-Smirnov two-sample test is applied:

KS_statistic, p_value = ks_2samp(baseline_times[device_id], recent_times) ...(16)

A p_value below a configurable threshold (preferred: 0.01, corresponding to 99% confidence) indicates that the observed device response timing is inconsistent with the registered hardware fingerprint for the claimed device identifier — a device impersonation indicator.

6.3.4 Graph-Based Lateral Movement Detection

The normal communication topology of the substation network is represented as a directed graph G_normal = (V, E_normal) with devices as nodes and observed communication sessions as directed edges. Engine (10) continuously maintains a live communication graph G_live and generates an immediate alert upon detection of any edge e in G_live that is absent from E_normal — a new communication relationship between devices that have no history of direct communication. Such a new edge is a reliable indicator of lateral movement by an attacker who has compromised one device and is attempting to reach another.

6.3.5 Replay Attack Detection

Engine (10) maintains a rolling cache of SHA-256 hashes of recent command payloads over a configurable window (preferred: 300 seconds). Detection of a payload hash previously observed in the cache generates an immediate high-confidence replay alert. Additionally, command sequence numbers and inter-arrival time uniformity are monitored: a coefficient of variation of inter-arrival times below a threshold (preferred: 0.05) indicates scripted replay rather than human operator-generated commands.

6.3.6 Honeypot Decoy Devices

The preferred embodiment of Engine (10) includes two or more virtual industrial controller decoy devices integrated into the substation network with device identifiers and network addresses that are plausible to an attacker but are absent from any legitimate SCADA configuration. Any communication directed to a decoy device by any source on the network generates an immediate CRITICAL threat alert with an anomaly score of 1.0 and a false positive rate of exactly zero, since no legitimate operational communication is configured to decoy devices.

6.3.7 Man-in-the-Middle Detection by RTT CUSUM

Engine (10) measures the round-trip time (RTT) for each SCADA communication session and maintains a per-session baseline RTT distribution. A Cumulative Sum (CUSUM) change detection algorithm is applied to the RTT time series:

$$C_t^+ = \max(0,\ C_{t-1}^+ + z_t - k) \quad \text{[upper CUSUM]} \quad ...(17)$$

$$C_t^- = \min(0,\ C_{t-1}^- + z_t + k) \quad \text{[lower CUSUM]} \quad ...(18)$$

where $z_t = (RTT_t - mu_{baseline}) / sigma_{baseline}$ is the normalised RTT deviation; k is the allowance parameter (preferred: 1.0 standard deviations); and an alert is generated when either $|C_t^+|$ or $|C_t^-|$ exceeds the decision boundary h (preferred: 5.0). A MITM proxy insertion typically adds 0.5 to 3 milliseconds of latency, which this CUSUM test detects within seconds of proxy insertion.

6.4 Valkyrie — Multi-Domain Fusion and Response Engine (40)

Referring to FIG. 4, the Valkyrie engine (40) receives anomaly scores from Dvvalin (10) and AEthelred (20) — designated S_D and S_A respectively — and computes a unified threat score S_T using confidence-weighted adaptive fusion:

$$w_A = w_{base_A} * conf_A \quad \text{[confidence-adjusted physics engine weight]} \quad ...(19)$$

$$w_D = w_{base_D} * conf_D \quad \text{[confidence-adjusted network engine weight]} \quad ...(20)$$

$$w_{A_norm} = w_A / (w_A + w_D) \quad \text{[normalised physics weight]} \quad ...(21)$$

$$w_{D_norm} = w_D / (w_A + w_D) \quad \text{[normalised network weight]} \quad ...(22)$$

$$S_T = w_{A_norm} * S_A + w_{D_norm} * S_D \quad \text{[weighted fusion score]} \quad ...(23)$$

A corroboration bonus is applied when both engines independently detect anomalous conditions:

$$\text{If } S_A > 0.5 \text{ AND } S_D > 0.5: \quad S_T = \min(1.0,\ S_T * c_{corr}) \quad ...(24)$$

where c_{corr} is the corroboration multiplier (preferred default: 1.25). The corroboration bonus reflects that simultaneous anomaly detection across independent physical domains provides substantially stronger evidence of a genuine cyber-physical attack than either signal alone, and reduces false positive rates by requiring cross-domain corroboration for high-confidence alerts.

The unified threat score S_T is classified into four threat levels:

Threat Level	Score Range	Automated Response Actions
CRITICAL	$S_T > 0.80$	Isolate affected buses; alert grid operator; capture full network forensics; generate CERT-In incident report; escalate to NCIIPC if national infrastructure

Threat Level	Score Range	Automated Response Actions
HIGH	0.65 < S_T <= 0.80	Alert security analyst; switch affected buses to backup sensor readings; enable enhanced monitoring; prepare isolation for operator confirmation
MEDIUM	0.45 < S_T <= 0.65	Alert grid operator; flag affected meters for field inspection; increase monitoring polling frequency
LOW	0.25 < S_T <= 0.45	Log event; increase monitoring sensitivity; no automatic operational action

6.5 Brynhildr — Thermodynamic Side-Channel Security Engine (30)

The Thermodynamic Side-Channel Security Engine (30) monitors the thermal and electromagnetic emission signatures of grid field hardware — distribution transformers, PLCs, RTUs, and IEDs — against expected thermodynamic profiles derived from the known power consumption model of each device under its current monitored operating conditions.

Physical modifications to device firmware, installation of hardware implants, or connection of hidden loads alter the power consumption profile and thermal emission pattern of affected devices in ways that are detectable through comparison against baseline thermodynamic profiles. The expected thermal state T_expected at time t is modelled as:

$$T_expected(t) = T_ambient(t) + k * P_device(t) \quad ...(25)$$

where T_ambient(t) is the ambient temperature, k is a device-specific thermal resistance coefficient, and P_device(t) is the expected device power consumption computed from the physical load model. A deviation between T_measured and T_expected exceeding a configured threshold constitutes a thermodynamic anomaly alert.

In the preferred embodiment, Engine (30) employs: infrared thermal imaging sensors co-located with field hardware; and electromagnetic emission sensors (looped wire probes or near-field antenna elements) for EM side-channel signature monitoring. Engine (30) is described herein and claimed in its full embodiment; detailed hardware implementation specifications are the subject of a related application.

6.6 Worked Examples

Note: *The following worked examples illustrate operation of the preferred embodiment on a simulated IEEE 33-bus distribution network with load parameters scaled to represent a typical Indian urban distribution feeder (nominal voltage 11 kV / 415 V, 50 Hz system frequency, T&D losses approximately 20% of feeder injection, consistent with CESC Mumbai distribution topology). All numerical values are representative of the described configuration.*

Example 1 — False Data Injection Attack Detection

The target network is modelled as a 33-bus radial distribution feeder. During a 48-hour normal operation training period, the AEthelred Module 3 PINN was trained for 20,000 epochs, converging to a physics residual L_physics of 0.00083 in normalised per-unit coordinates and a data fidelity loss L_data of 0.00041. The Neyman-Pearson detection

threshold tau was calibrated on a held-out 6-hour validation dataset at tau = 0.0024, yielding a false positive rate of 4.2% on the diurnal load cycle.

At time t = 0, a False Data Injection attack was injected by inflating the voltage magnitude readings at buses 3 and 5 by a factor of 1.15 (15% inflation), representing an attacker who has compromised the sensors at those buses and reports falsely elevated voltage readings to the SCADA state estimator. The network traffic at all times remained fully protocol-compliant and within normal behavioural bounds — no network-layer IDS would generate any alert.

The following observations were recorded:

Parameter	Normal Operation	Under FDI Attack	Amplification Ratio
Physics residual at bus 3 (pu)	0.00031	0.04876	157.3x
Physics residual at bus 5 (pu)	0.00028	0.04124	147.3x
Mean physics residual all buses (pu)	0.00029	0.00412	14.2x
KCL violation at bus 3 (pu)	0.00002	0.00387	193.5x
Valkyrie S_A score	0.041	0.923	—
Time to detection (seconds)	—	3.8	—
Threat level assigned	NORMAL	CRITICAL	—

The detection latency of 3.8 seconds represents the time from attack injection to Valkyrie CRITICAL alert generation, comfortably within the 4-second target. The 157-fold amplification of the physics residual at the attacked bus provides a very high signal-to-noise ratio, enabling reliable detection even with a conservative detection threshold.

Example 2 — Electricity Theft Detection

In the same 33-bus simulation, electricity theft was simulated by disconnecting the smart meter at bus 7 from the billing network while the consumer continued to draw power from the distribution transformer. The stolen power was set to 0.5 MW, representing a consumer drawing approximately five times the average residential consumption unmetered.

Parameter	Normal Operation	Under Theft Attack	Change
Feeder input power P_feeder_in (MW)	4.82	5.32	+10.4%
Sum of meter readings SUM(P_meter_k) (MW)	3.98	3.98	0% (theft invisible to meters)
Estimated technical losses P_loss (MW)	0.78	0.86	+10.3%
NTL ratio	0.021 (2.1%)	0.089 (8.9%)	+7.8 percentage points
NTL detection threshold	0.05 (5%)	—	—
Slow-window detection time	—	48 seconds	—

Parameter	Normal Operation	Under Theft Attack	Change
Threat level assigned	NORMAL	MEDIUM	—

The NTL ratio of 8.9% substantially exceeded the 5% detection threshold. The theft was detected within the 150-reading slow window (48 seconds at 2-second polling interval). The recommended action generated was: dispatch field inspection team to bus 7; flag bus 7 meters in the billing system for physical audit. The detection is independent of the integrity of the bus 7 meter — tampering with the meter has no effect on the feeder-level energy balance computation.

Example 3 — SCADA Command Injection with Multi-Domain Corroboration

An unauthorised Write Single Coil command (Modbus Function Code 5) was injected targeting PLC Unit ID 2, which controls a circuit breaker on the 11 kV busbar. PLC Unit ID 2 had never issued or received a write command during the 48-hour training period — its behavioural profile contained only Read Holding Registers (Function Code 3) at regular 500-millisecond polling intervals.

Detection Layer	Response	Score / Confidence
Dvvalin Rule Layer	Write to critical device (FC 5 to Unit ID 2) — immediate flag	S_D rule = 0.90
Dvvalin Behavioural Profile	FC 5 absent from trained profile for Unit ID 2	S_D profile = 0.82
Dvvalin Combined S_D	Rule + profile fusion	S_D = 0.91
AEthelred Module 4	Voltage spike at buses 3-5 following command (physical effect)	S_A = 0.73
Valkyrie S_T (pre-corroboration)	w_D=0.45, w_A=0.55, weighted sum	S_T = 0.82
Valkyrie corroboration bonus	Both S_A > 0.5 and S_D > 0.5: S_T * 1.25	S_T = 1.00 (capped)
Threat level	CRITICAL	Confidence: 0.97
MITRE ATT&CK mapping	T0855 — Unauthorized Command Message	—
Time to detection (seconds)	1.4 seconds	—

6.7 Deployment Configurations

Referring to FIG. 7, the system of the present invention supports three deployment configurations:

(a) Cloud-Connected Configuration: all four engines are deployed on a server at the substation or in a cloud computing environment. Real-time sensor data is ingested via a network tap. The PINN model is executed on a GPU-enabled inference server. Wazuh and OpenSearch provide SIEM functionality with

centralised alert aggregation across multiple substations. A REST API conforming to the OpenAPI 3.0 specification provides integration with existing SCADA Distribution Management Systems.

(b) Edge-Deployed Configuration: the PINN model is exported to the ONNX (Open Neural Network Exchange) format and converted to TensorRT optimised inference format for execution on an NVIDIA Jetson Nano embedded computing device (approximately 5 watts power consumption) co-located at the substation. This configuration operates without any continuous network connectivity — all detection, alerting, and logging functions are performed locally. Events are synchronised to a central SIEM when connectivity is available. This configuration is specifically designed for rural Indian substations with unreliable internet connectivity.

(c) Federated Configuration: a plurality of edge-deployed instances, one per substation of a distribution utility, periodically contribute locally computed PINN model parameter updates to a central aggregation server using the federated averaging (FedAvg) algorithm. The central server aggregates updates into an improved global model and distributes it to all substation instances. This federated learning approach enables all substations to benefit from attack events detected at any substation without any exchange of raw sensor data, preserving the operational confidentiality of each substation.

7. CLAIMS

Drafting note (for patent agent): *All method claims are drafted to recite a technical effect on a physical power distribution network and avoid pure mathematical method characterisation, pursuant to the CRI Guidelines 2017 and Section 3(k) of the Patents Act 1970. System claims recite hardware components together with their functional configuration. Claims 1-4 are independent; Claims 5-24 are dependent. Agent should verify claim language against the latest IPO examination guidelines before filing.*

What is claimed is:

INDEPENDENT CLAIMS

1. A cyber-physical intrusion detection and response system for securing a physical electric power distribution network against cyber-physical attacks, the system comprising: a network monitoring unit (21) comprising one or more processors and network interface hardware configured to capture and parse industrial protocol traffic on the distribution network's operational technology communication network, the monitored protocols including at least one of Modbus TCP, DNP3, IEC-61850 GOOSE, IEC-61850 MMS, and OPC-UA, and to extract therefrom a plurality of physical state readings comprising voltage magnitude, voltage angle, active power, and reactive power at each monitored network bus; a physics constraint processor (22) comprising one or more processors configured to solve the full nonlinear AC power flow equations for the network topology and compute expected physical state values h(x_expected) at each bus, and to compute a Non-Technical Loss ratio as specified in equation (3) for electricity theft detection; a Physics-Informed Neural Network processor (23) comprising one or more processors running a neural network trained with the composite loss function of equations (6)-(9) embedding AC power flow governing equations as differentiable constraints, configured to produce physics-consistent predicted state values u_hat(x,t); a physics residual anomaly processor (24) comprising one or more processors configured to compute, for each monitored bus, a False Data Injection residual r(z) according to equations (10)-(11) and to generate an anomaly detection signal when said residual exceeds a calibrated threshold, the technical effect of said detection being prevention of incorrect protective relay operation and incorrect load dispatch caused by falsified sensor measurements; a threat classification processor (25) comprising one or more processors configured to classify detected anomalies by type and to perform causal root-cause analysis distinguishing cyber attacks from physical equipment faults; and a fusion and response processor (40) comprising one or more processors configured to combine anomaly signals from the physics residual anomaly processor (24) and from a network intrusion detection engine (10) using the confidence-weighted fusion of equations (19)-(24), and to execute automated protective response actions on the physical distribution network corresponding to the classified threat level.

2. A computer-implemented method for protecting a physical electric power distribution network from False Data Injection attacks, the method producing the technical effect of detecting physically inconsistent sensor measurements in the distribution network before they cause incorrect state estimation, erroneous protective relay operation, or incorrect load dispatch, the method comprising: receiving, by a processor, a plurality of observed sensor readings z_observed comprising voltage magnitudes, voltage angles, active powers, and reactive powers at a plurality of network buses; computing, by said processor operating a Physics-Informed Neural Network trained with the composite loss function

L_total = lambda_d*L_data + lambda_p*L_physics + lambda_b*L_boundary wherein L_physics encodes Kirchhoff's Current Law and AC power flow equation residuals, a physics-consistent predicted state u_hat(x,t) for each bus; computing, by said processor executing an AC power flow solver on the network topology, a physics oracle expected state h_oracle(x) for each bus; computing a False Data Injection residual r(z) = |z_observed - h(x_expected)| wherein h(x_expected) = alpha*u_hat(x,t) + (1-alpha)*h_oracle(x); applying a statistical test to said residual to produce an anomaly score; and generating, when said anomaly score exceeds a Neyman-Pearson calibrated threshold, an alert identifying the affected network bus, the physical law violated, and the recommended protective action to be taken on the physical distribution network.

3. A computer-implemented method for detecting electricity theft in a physical smart-metered electric power distribution network, the method producing the technical effect of identifying unmetered energy consumption in the distribution network thereby enabling recovery of Non-Technical energy Losses caused by meter tampering or bypass, the method comprising: receiving, by a processor, a feeder-level power injection measurement P_feeder_in from a substation-level metering point; receiving a plurality of individual customer smart meter consumption readings P_meter_k for each of M smart meters on said feeder; computing estimated technical line losses P_estimated_loss from the feeder network impedance model and measured current values; computing a Non-Technical Loss ratio NTL_ratio = (P_feeder_in - SUM_k(P_meter_k) - P_estimated_loss) / P_feeder_in, wherein said computation is independent of the firmware and hardware integrity of any individual smart meter; detecting electricity theft when said NTL_ratio exceeds a configurable threshold over a rolling multi-reading detection window; and generating an alert identifying the affected feeder segment and recommending physical inspection of meter installations in said segment, the alert including an estimated stolen power quantity computed as P_stolen = NTL_ratio * P_feeder_in.

4. A computer-implemented method for multi-domain anomaly fusion in a cyber-physical intrusion detection system for a physical electric power distribution network, the method producing the technical effect of reducing false positive alerts from normal load transients while maintaining high detection sensitivity for coordinated cyber-physical attacks by requiring independent corroborating evidence from multiple physical observation domains, the method comprising: receiving a physics-domain anomaly score S_A computed by a Physics-Informed Neural Network anomaly detection engine monitoring electrical physics consistency of sensor measurements; receiving a network-domain anomaly score S_D computed by a SCADA network intrusion detection engine monitoring industrial protocol traffic; computing respective confidence-adjusted weights w_A = w_base_A * conf_A and w_D = w_base_D * conf_D and normalising said weights; computing a weighted fusion score S_T = w_A_norm * S_A + w_D_norm * S_D; applying a corroboration bonus multiplier c_corr to S_T when both S_A and S_D independently exceed a corroboration threshold, amplifying the fusion score when independent physical domains simultaneously detect anomalous conditions; classifying S_T into threat levels; and executing automated protective response actions on the physical distribution network corresponding to the classified threat level, including at minimum: generating an explanatory alert with SHAP feature attribution and MITRE ATT&CK for ICS mapping.

DEPENDENT CLAIMS

5. The system of claim 1, wherein the Physics-Informed Neural Network processor (23) implements a Physics-Informed Graph Attention Network representing the power network as a directed graph with buses as nodes and transmission lines as edges,

wherein message passing between neighbouring nodes is weighted by learned attention coefficients, and the physics residual loss L_physics is computed per node and aggregated across the graph, producing a detection sensitivity improvement of approximately ten to fifteen percent for FDI attacks affecting multiple electrically adjacent buses compared to the feedforward PINN embodiment.

6. The system of claim 1, wherein the Physics-Informed Neural Network processor (23) implements Bayesian uncertainty quantification using Monte Carlo Dropout, performing N stochastic forward passes with dropout active at inference time to produce a predictive mean mu_hat and a per-output predictive standard deviation sigma_hat, and wherein the physics residual anomaly processor (24) generates an anomaly signal only when the observed measurement deviates from mu_hat by more than a configurable multiple of sigma_hat, substantially reducing false positive rates under legitimate load transients.

7. The system of claim 1, wherein the physics residual anomaly processor (24) implements a multi-timescale detection architecture comprising a fast detection window of N_fast consecutive readings sensitive to transient FDI attacks and a slow detection window of N_slow consecutive readings sensitive to gradual electricity theft and slow-ramp load manipulation, wherein N_slow is at least ten times N_fast.

8. The system of claim 1, wherein the threat classification processor (25) performs causal root-cause analysis by applying Granger causality testing to residual time series across multiple monitored buses, distinguishing cyber attacks characterised by causal propagation from a single source bus from physical equipment faults characterised by simultaneous uncorrelated anomalies across electrically coupled buses and from sensor hardware failures characterised by isolated single-bus anomaly without spatial correlation to adjacent buses.

9. The system of claim 1, further comprising a network intrusion detection engine (10) configured to monitor industrial protocol traffic including at least Modbus TCP, DNP3, IEC-61850 GOOSE, IEC-61850 MMS, and OPC-UA; to maintain per-device behavioural profiles comprising function code frequency distributions, register access ranges, and inter-arrival time distributions; and to generate network-domain anomaly scores for input to the fusion and response processor (40).

10. The system of claim 9, wherein the network intrusion detection engine (10) performs device identity authentication by: recording baseline response time distributions for each registered device identifier during a training period; and applying a two-sample Kolmogorov-Smirnov statistical test to compare observed response time distributions against registered baselines, generating a device impersonation alert when the test p-value falls below a configured significance threshold.

11. The system of claim 9, wherein the network intrusion detection engine (10) represents the normal substation communication topology as a directed graph and generates an immediate high-confidence alert upon detection of any communication session between two devices that have no record of prior direct communication in the baseline graph, indicating lateral movement by an attacker who has compromised one device and is accessing another.

12. The system of claim 9, wherein the network intrusion detection engine (10) integrates one or more virtual industrial controller decoy devices having plausible device identifiers and network addresses present on the operational technology network but absent from all legitimate SCADA configuration documents, wherein any communication directed to a decoy device generates an immediate anomaly score of 1.0 with a false positive rate of exactly zero.

13. The system of claim 9, wherein the network intrusion detection engine (10) detects Man-in-the-Middle proxy insertion by applying the two-sided CUSUM change detection algorithm of equations (17) and (18) to per-session round-trip time measurements, generating an alert when the cumulative sum statistic exceeds the decision boundary h.

14. The system of claim 9, wherein the network intrusion detection engine (10) detects replay attacks by maintaining a rolling cryptographic hash cache of command payloads over a configurable time window and generating an immediate replay alert upon detection of a payload hash previously observed in said cache.

15. The system of claim 1, further comprising a thermodynamic side-channel security engine (30) configured to monitor thermal emission signatures of field hardware using infrared thermal sensors co-located with distribution transformers and programmable logic controllers, compare measured thermal values against expected thermal profiles computed from equation (25), and generate a thermodynamic-domain anomaly score for input to the fusion and response processor (40).

16. The method of claim 2, further comprising computing voltage stability L-indices at load buses according to equations (4) and (5) and generating operational voltage collapse warning alerts when any L-index exceeds 0.5, independently of any cyber attack detection, thereby providing grid health monitoring functionality in addition to the cyber-physical security function.

17. The method of claim 2, wherein the Physics-Informed Neural Network is pre-trained on an IEEE standard power network test case and fine-tuned on target network operational data using transfer learning with selective layer freezing, wherein the first sixty percent of network layers encoding universal AC power flow physics are frozen during fine-tuning and the remaining layers encoding network-specific topology patterns are updated, reducing the required target-network training dataset size from over fifty thousand samples to approximately five thousand samples.

18. The method of claim 2, wherein the Physics-Informed Neural Network is continuously updated by an online adaptive learning mechanism that fine-tunes network parameters on a rolling buffer of recent confirmed-normal operational readings using an exponential forgetting factor that weights recent observations more heavily than older observations, preventing detection accuracy degradation from seasonal and long-term load pattern drift.

19. The method of claim 4, wherein the SHAP feature attribution of the generated alert decomposes the anomaly score into per-feature contributions using SHapley Additive exPlanations values, identifying for each alert the dominant contributing feature by

name, the direction and magnitude of its contribution, and the percentage of the total anomaly score attributable to it, providing actionable explanations to grid operators.

20. The method of claim 4, further comprising automatically mapping each classified threat event to a corresponding tactic and technique in the MITRE ATT&CK for Industrial Control Systems framework and automatically generating a structured incident report conforming to CERT-In mandatory notification format including attack classification, timeline, affected network components, MITRE ATT&CK mapping, and recommended response actions.

21. The method of claim 2, further comprising classifying the detected anomaly into at least the following categories using a trained classification model: False Data Injection; Electricity Theft; Load Manipulation; SCADA Command Injection; Device Impersonation; Coordinated Cyber-Physical Attack; and Physical Equipment Fault or Sensor Hardware Failure; wherein the classification is used to select a category-specific automated response action from a configurable response playbook.

22. The system of claim 1, wherein the fusion and response processor (40) generates a tamper-evident hash-chain audit log wherein each log entry contains a cryptographic hash of its own content concatenated with the hash of the preceding entry, such that any retrospective modification of a log entry is detectable by recomputing the hash chain, satisfying the evidence preservation requirements of Section 79A of the Information Technology Act, 2000.

23. The system of claim 1, wherein the Physics-Informed Neural Network processor (23) is deployed in an edge-computing configuration on an embedded processing device co-located with the substation, the neural network model having been exported to an optimised inference format and executed by a hardware-accelerated inference engine, the system operating without dependence on external network connectivity, enabling deployment at substations with unreliable or absent internet connectivity.

24. The system of claim 23, wherein a plurality of instances of the system are deployed at respective substations of a distribution utility, and model parameters are periodically updated using a federated learning protocol that aggregates locally computed parameter update vectors from each substation instance without exchange of raw sensor data, enabling collective improvement of anomaly detection capability across all substations of the utility while preserving the operational data confidentiality of each individual substation.

8. ABSTRACT OF THE DISCLOSURE

A cyber-physical intrusion detection and response system for electric power distribution networks is disclosed, comprising four cooperating detection engines monitoring the cyber, electrical physics, and thermodynamic domains simultaneously. The system employs a Physics-Informed Neural Network (PINN) trained with a composite loss L_total = lambda_d*L_data + lambda_p*L_physics + lambda_b*L_boundary that embeds full nonlinear AC power flow governing equations as differentiable constraints. At inference, a False Data Injection residual r(z) = |z_observed - h(x_expected)| is computed using a dual oracle combining the PINN prediction and an AC power flow physics oracle, detecting FDI attacks that are provably undetectable by linearised chi-squared methods. Electricity theft is detected via a Non-Technical Loss ratio computed from feeder-level and meter-level power measurements, immune to individual meter tampering. SCADA network attacks are detected by a multi-protocol network IDS engine with per-device behavioural profiling, hardware timing fingerprinting, graph-based lateral movement detection, and honeypot decoy devices. A multi-domain fusion engine combines physics and network anomaly scores with adaptive confidence weighting and corroboration bonus. All alerts include SHAP explainability, MITRE ATT&CK for ICS mappings, and automated CERT-In incident reports. The system supports cloud-connected, edge-deployed offline, and federated multi-substation deployment configurations.

Keywords: cyber-physical security; power grid protection; Physics-Informed Neural Networks; False Data Injection detection; electricity theft detection; SCADA intrusion detection; multi-domain anomaly fusion; industrial control system security; smart grid.

DECLARATION BY THE APPLICANT / INVENTOR

I/We hereby declare that:

(a) I am/We are the true and first inventor(s) of the invention described in this Complete Specification.

(b) The invention as disclosed in the Complete Specification filed herewith has not been claimed in any other complete specification filed in India.

(c) No patent or patent application for the same or substantially the same invention has been filed by me/us or with my/our knowledge and consent in any country other than India before the date of this application, except as may be disclosed on Form 3 filed herewith.

Signature of Applicant/Inventor: _____

Name (in full): [FULL LEGAL NAME]

Date: _____ Place: _____

THE PATENTS ACT, 1970

FORM 3

(See Section 8 and Rule 12)

STATEMENT AND UNDERTAKING BY THE APPLICANT

Application No.: _____ Date of Filing: _____

Title of Invention: A MULTI-DOMAIN CYBER-PHYSICAL INTRUSION DETECTION AND RESPONSE SYSTEM FOR ELECTRIC POWER DISTRIBUTION NETWORKS USING PHYSICS-INFORMED NEURAL NETWORKS AND MULTI-LAYER ANOMALY FUSION

STATEMENT UNDER SECTION 8(1) — FOREIGN APPLICATIONS

I/We the applicant(s) hereby state that:

[ ] I/We have NOT filed or made or caused to be filed or made any application for patent in respect of the same or substantially the same invention in any country outside India before the date of filing of this application in India.

[ ] I/We have filed or caused to be filed application(s) for patent in respect of the same or substantially the same invention in the following countries outside India:

Country	Application Number	Date of Filing	Status
Not Applicable	—	—	—

UNDERTAKING UNDER SECTION 8(2)

I/We the applicant(s) hereby undertake that up to the date of grant of patent by the Indian Patent Office, I/We shall keep the Controller informed in writing, from time to time, of detailed particulars as required under Section 8(1) of the Patents Act, 1970, in respect of every application for patent filed in any country outside India in respect of the same or substantially the same invention within six months from the date of filing of each such foreign application.

Signature: _____ Date: _____

Name: [FULL LEGAL NAME] Place: _____

THE PATENTS ACT, 1970

FORM 5

(See Section 7(2) and Rule 13(6))

DECLARATION AS TO INVENTORSHIP

Application No.: _____ Date of Filing: _____

Title of Invention: A MULTI-DOMAIN CYBER-PHYSICAL INTRUSION DETECTION AND RESPONSE SYSTEM FOR ELECTRIC POWER DISTRIBUTION NETWORKS USING PHYSICS-INFORMED NEURAL NETWORKS AND MULTI-LAYER ANOMALY FUSION

I/We the applicant(s) do solemnly declare that:

(a) The true and first inventor(s) of the invention described in the Complete Specification is/are the person(s) set out below.

(b) No other person is, to my/our knowledge, the inventor or a joint inventor of the invention described in this specification.

Inventor No.	Full Legal Name	Nationality	Address
1	[FULL LEGAL NAME]	Indian	[Complete Address with PIN Code]

I/We make this declaration conscientiously believing the same to be true.

Verified at _____ on this _____ day of _____ 2026.

Signature: _____ Date: _____

Name: [FULL LEGAL NAME]

Designation: Applicant / Inventor

IPO FILING CHECKLIST

Important: *All items below must be submitted together as a single filing package at the Indian Patent Office. Do not submit the Complete Specification (Form 2) without Form 1. The filing date is the date on which all mandatory documents and the prescribed fee are received by the IPO.*

#	Document	Form No.	Mandatory?	Status
1	Request for Grant of Patent	Form 1	YES	[ ] Included
2	Complete Specification with Claims and Abstract	Form 2	YES	[ ] Included
3	Statement and Undertaking re Foreign Applications	Form 3	YES	[ ] Included
4	Declaration as to Inventorship	Form 5	YES (individual)	[ ] Included
5	Technical Drawings (FIG. 1 through FIG. 8)	As per Rule 32	YES (referenced in spec)	[ ] Attach separately
6	Power of Attorney in favour of Patent Agent	Form 26	If using agent	[ ] If applicable
7	Filing Fee — Individual Applicant (e-filing)	₹1,600	YES	[ ] Pay online at ipindia.gov.in
8	Provisional Application Reference	Prov. App. No.	If claiming priority	[ ] Quote in Form 1
9	Sequence Listing (if biological material)	Annex	Not applicable	[ ] N/A

Fee Category	e-Filing (Reduced)	Physical Filing
Individual Applicant (Natural Person)	₹1,600	₹4,000
Startup (DPIIT Recognised)	₹800	₹2,000
Small Entity	₹4,000	₹10,000
Large Entity / Company	₹8,000	₹20,000

Note: DPIIT startup recognition (available at startupindia.gov.in) qualifies for 80% fee reduction and is strongly recommended before filing. The examination request fee (Form 18) must be filed separately within 48 months of the filing date.